



Open-source liquid chromatography-mass spectrometry data processing tools for RNA modification analysis

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ABSTRACT

RNA modifications have attracted increasing attention due to their importance in biological functions and the development of RNA therapeutics. Over the past decades, >170 biochemical modifications of RNA have been identified, which play physiological and pathological roles. Besides those naturally occurring modifications, synthetic chemical modifications are important to RNA-based therapeutics. Currently, liquid chromatography-mass spectrometry (LC-MS) stands as the critical technique capable of directly and comprehensively characterizing a large variety of chemical modifications in RNAs. However, the reliance on commercial software for LC-MS data processing can restrict accessibility, flexibility, and customization for researchers. Open-source tools provide a valuable alternative by offering cost-effective, transparent, and adaptable solutions. In this review, we examine LC-MS methodologies and open-source data processing tools for RNA modification analysis, highlighting their strengths, limitations, and suitability for specific research applications. We also discuss future advancements in LC-MS approaches, including the integration of artificial intelligence and machine learning to enhance data analysis.

1. Introduction

Naturally occurring RNA molecules in almost all organisms contain a variety of chemically modified nucleosides [1]. To date, >170 post-transcriptional modifications of RNAs that are naturally presented have been identified and reported in databases such as MODOMICS [2] and RNAMDB [3]. In addition to well-characterized marks like m⁶A and Ψ, recent studies have identified emerging modifications such as N4-acetylcytidine (ac⁴C) [4], 5-hydroxymethylcytosine (hm⁵C) [5], and N3-methylcytidine (m³C) [6], whose functional significance is still being unraveled. These modifications are prevalent in different RNA species, ranging from messenger RNAs (mRNAs) to non-coding RNAs such as ribosomal RNAs (rRNAs), transfer RNAs (tRNAs), and microRNAs [7]. Among these, tRNAs have the highest percentage of chemical modifications, while mRNAs exhibit much lower levels of modifications [8].

Naturally occurring RNA modifications have been reported to play an important role in RNA structure folding and stability [9], base pairing

[10], regulation of gene expression [11], and cell development [12]. Writer enzymes (e.g., methyltransferase-like 3 (METTL3) for m⁶A), erasers (e.g., fat mass and obesity-associated protein, FTO), and readers (e.g., YTH domain-containing family protein 1, YTHDF1) dynamically regulate RNA modifications to fine-tune gene expression [13]. Writers add chemical marks (like m⁶A) to RNA, influencing splicing, stability, and translation. Erasers remove these marks, reversing their effects, while readers recognize them to mediate downstream processes such as RNA decay or protein synthesis. This dynamic interplay allows cells to rapidly adapt to environmental changes, influencing development, stress responses, and disease states like cancer. By modulating RNA fate and function, these mechanisms shape cellular physiology with precision. For example, N⁶-methyladenosine (m⁶A), the most abundant modification of mRNA [14], regulates almost all aspects of mRNA metabolism, from expression and pre-mRNA processing in the nucleus to translation and mRNA decay in the cytoplasm [15]. Pseudouridine (Ψ), one of the most abundant modifications in tRNAs and rRNAs [16], is

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essential for tRNA structure stabilization and rRNA folding [14]. Moreover, this modification can affect both the speed and fidelity of mRNA translation [17]. Aberrant RNA modifications can lead to human illnesses, including neurological diseases, metabolic disorders and cancers [18].

In addition to being present in naturally occurring RNA molecules, chemical modifications are incorporated in RNA-based therapeutics and mRNA vaccines to enhance stability, translation efficiency, and immune response. Optimization of mRNAs by incorporating 5-methylcytidine (m^5C) and N1-methylpseudouridine ($m^1\Psi$) was found to minimize immunogenicity and increase translation efficiency, as seen in COVID-19 vaccines [19,20]. Additionally, optimized 5' cap structures (a modified guanine nucleotide added to the front of mRNA. It is essential for ribosome binding to initiate translation and to protect the transcript from degradation) and poly(A) tails (a long chain of adenine nucleotides added to the 3' end which promotes mRNA stability and aids in export from the nucleus and translation) enhance mRNA stability and translation efficiency [21,22]. In siRNA therapeutics, 2'-O-methyl (2'-OMe), 2'-fluoro (2'-F), and phosphorothioate (PS) modifications improve stability and target binding [23]. These chemical modifications are critical for the success of RNA-based drugs and vaccines. SynONIM is a comprehensive database of synthetic oligonucleotide modifications [24], which contains the information of 80 modifications or impurities observed in the synthetic oligonucleotides.

Analysis of RNA modifications is crucial to the understanding of the functions of naturally occurring RNA modifications and the development of RNA therapeutics. Liquid chromatography-mass spectrometry (LC-MS) has emerged as a cornerstone technology in the analysis of RNA modifications, offering unparalleled specificity and versatility for the identification and quantification of diverse RNA modifications [25]. Its ability to resolve complex mixtures and detect low-abundance RNA modifications has significantly advanced the understanding of the epitranscriptome as well as the development of RNA therapeutics [26]. However, the vast and complex datasets generated by LC-MS necessitate robust, efficient, and accessible data analysis tools to obtain meaningful biological and analytical insights. While commercial software such as BioPharma Finder (Thermo Fisher Scientific), ProMass (Novatia), OligoQuest (Bruker Daltonics) and PMI-Byos Oligo workflow (Protein Metrics) provide powerful solutions for LC-MS data processing, their proprietary nature can limit accessibility and flexibility for researchers. Compared with the commercial software, the open-source tools allow modification of parameters for rare or newly discovered RNA modifications, which may not be well-supported in commercial software. Additionally, they often integrate the latest computational approaches include customizable search algorithms and statistical validation strategies. Furthermore, open-source tools can be implemented in larger analytical framework (e.g., OpenMS), which allows users to build tailored workflows by combining existing tools or writing new ones in

C++/Python. Therefore, open-source data processing tools offer customizable, transparent, and cost-effective alternatives. This review summarizes existing LC-MS based methods for RNA modification analysis and focuses on the open-source software as a critical complement to commercial solutions, highlighting its role in advancing research in the area of RNA modifications.

2. LC-MS analysis of RNA modifications

LC generally reduces sample complexity by separating molecules and removing salts from RNA formulation buffers prior to MS detection. Detection of RNA modifications often relies on high-resolution MS, such as time-of-flight (TOF), Orbitrap, and Fourier-transform ion cyclotron resonance (FT-ICR) mass spectrometers. These instruments provide the mass resolving power and accuracy needed to determine the exact m/z values of modifications. High-sequence coverage can be achieved by using product ions (Fig. 1), which are generated from MS detection mode with collision-, electron- or photo-dissociated experiments. RNA modification analysis can be generally divided into top-down and bottom-up approaches (Fig. 2). The main difference is that top-down analysis targets full-length RNA and reconstructs sequences using product ions generated by different MS dissociation methods, without requiring RNA hydrolysis. In contrast, bottom-up analysis involves hydrolyzing RNA into oligonucleotide fragments or individual nucleotide units for RNA modification detection. Given the complementary strengths of these techniques, the choice of LC-MS strategy depends on the specific analytical goals in the analysis of RNA modifications,

2.1. RNA mapping strategies

The mapping of small RNAs (≤ 100 nt), such as oligonucleotides [27], tRNA [28] and single guide RNA (sgRNA) [29], is amenable to top-down LC-MS analysis due to their relatively short length, which allows for direct characterization of modifications (Table 1). However, this approach is not applicable to the mapping of large RNA, e.g., mRNA and self-amplifying RNA (saRNA), could be up to 12,000 nt long [30]. Therefore, the bottom-up approach, which involves digestion of RNA into more manageable oligonucleotides is primarily applied [31–34]. This workflow enables characterization of RNAs based on the specific digestion products. Additionally, complete digestion of RNA to single nucleotides can enabling quantification of known RNA modifications with detection limits up to the attomole levels [35].

2.1.1. Top-down analysis of small RNA modifications

Because small RNA (e.g., oligonucleotide or siRNAs) sequences are relatively short, they are generally analyzed through top-down mass spectrometry methods. Top-down analysis of full-length RNA enables total mass analysis, identification of modifications, and localization

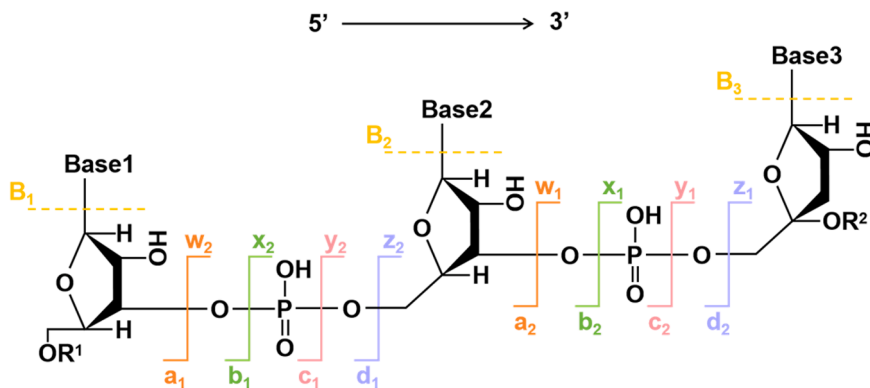


Fig. 1. The McLuckey nomenclature of RNA fragment ions. The product ions from the 5'-end of the molecule are a, b, c, and d ions, and the product ions from the 3'-end are w, x, y, and z ions.

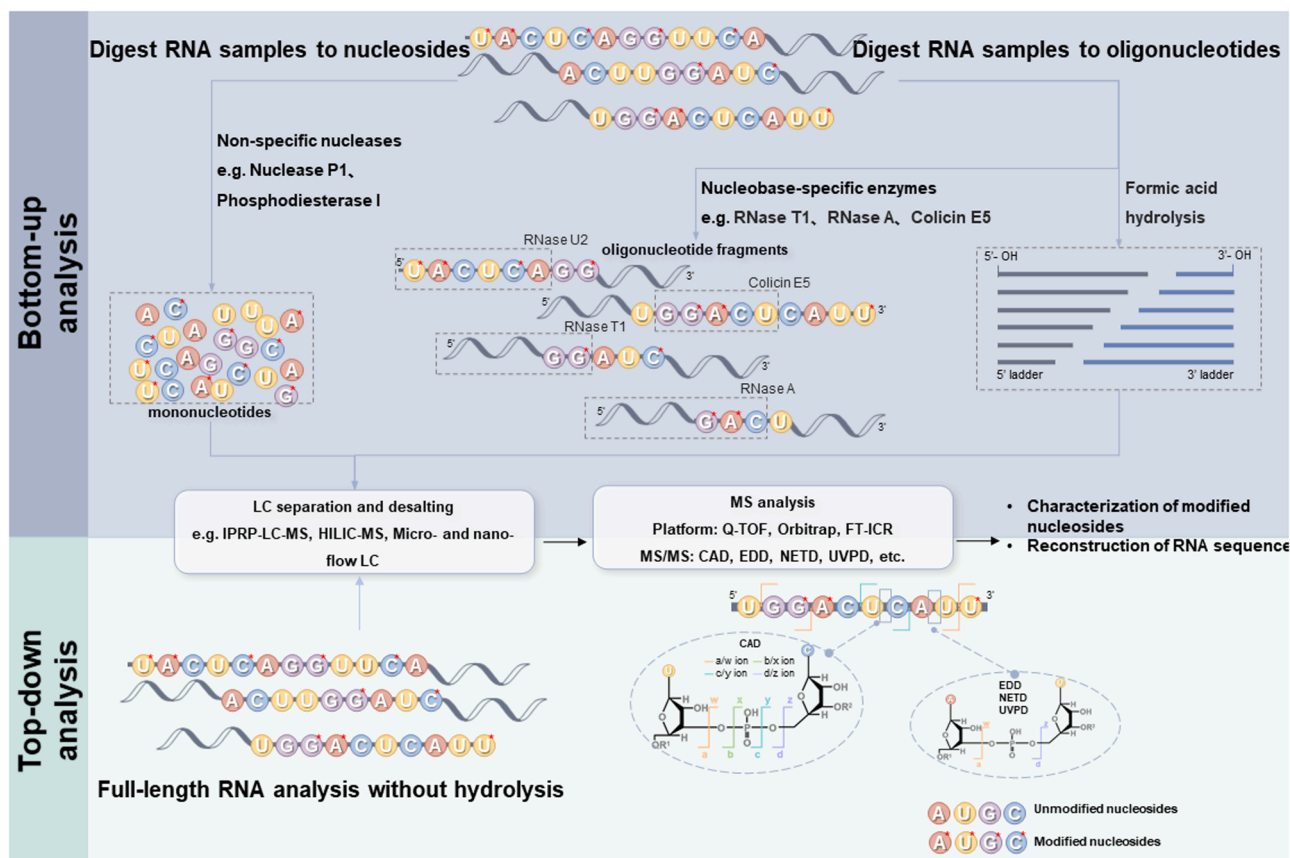


Fig. 2. Schematic overview of LC-MS-based strategies for mapping RNA modifications via top-down and bottom-up approaches.

within the sequence context [35]. The general protocol involves purifying the target RNA to near homogeneity, followed by high resolution mass spectrometry (HRMS) analysis, and specialized MS/MS fragmentation methods are applied to break RNA molecules into smaller ions for structural analysis. Then utilize these typical product ions to reconstruct RNA sequences.

The key advantages of top-down analysis include the elimination of laborious enzymatic digestion steps and the ability to provide nearly complete sequence and modification mapping coverage. However, this approach requires highly pure RNA samples, restricting its applicability to the analysis of complex mixtures. Additionally, interpreting the resulting spectra is challenging due to their high complexity, which highlights the need for the development of computational resources to enhance the accessibility and utility of top-down LC-MS strategies.

2.1.2. Bottom-up analysis for large RNA modifications using nuclease hydrolysis

Large RNAs (e.g., mRNAs) with greater length and molecular weight pose analytical challenges for full-length mass spectrometric characterization. Therefore, large RNAs are generally detected through bottom-up mass spectrometry methods, which involve hydrolytic digestion prior to detection. The bottom-up analysis of RNA modifications relies on enzymatic digestion of RNA into oligonucleotide or nucleosides (Table 1). Nonspecific nucleases, such as nuclease P1, nuclease S1 or benzonase (endonucleases) and phosphodiesterase I (an exonuclease), degrade RNA into mononucleotides, followed by dephosphorylation with bacterial alkaline phosphatase to generate free nucleosides [36,37]. This method is highly sensitive and allows for quantitative estimation of modified ribonucleosides [38]. However, information about the precise location of modifications in the RNA sequence is lost. In contrast, oligonucleotide-based analysis utilizes nucleobase-specific ribonucleases (RNase), such as RNase T1 (cleaves after guanines) [39],

RNase A (pyrimidines) [40], RNase U2 (purines) [41], cusativin (cleave Cp[U/A/G] dinucleotide combinations along with UpA and [A/U]pU), and MC1 (cleave the [A/U/C]pU dinucleotide bond) [42], to generate defined oligonucleotide sequences. This approach preserves sequence information and enables modification site mapping. However, its success depends on enzyme specificity, and incomplete digestion or overlapping cleavage patterns can complicate data interpretation. As reviewed by Maurer et al. [34], parallel digestion with different RNases can enhance full sequence coverage. For example, it was reported that an improved RNA sequence mapping was obtained with a combination of RNases T1, cusativin and MC1, with sequence coverage reached 75–100 % for tRNAs (70–90 nt) and >60 % for 2904 nt long 23S rRNA [43]. While nucleoside analysis is ideal for global modification quantification, oligonucleotide-based methods are essential for site-specific characterization, making both strategies complementary in RNA modification research.

The analysis of RNA modifications using LC-MS after ribonucleases enzymatic digestion presents several data processing challenges. First, the digestion process generates a complex mixture of oligonucleotide sequences with varying lengths, modifications, and charge distributions, making peak identification and annotation difficult. Additionally, overlapping cleavage patterns from different ribonucleases will complicate sequence reconstruction and modification site mapping. Moreover, the presence of isomeric and isobaric modifications will require LC separation and/or MS/MS fragmentation analyses to differentiate accurately.

2.1.3. Ladder sequencing using formic acid digestion

Szostak group demonstrated the LC-MS sequencing and modification mapping of RNA through sequence ladders generated by formic acid digestion [44]. Two complementary oligonucleotide sequences were generated from the cleavage of the RNA phosphodiester bond. In ion

Table 1

The enzymes used for RNA digestion in the analysis of RNA modifications.

Enzymes	Cleavage site	Product	Applications
RNase T1	3'-end of Guanosine and N2-methylguanosine (m2G) [39]	Oligonucleotides	Sequence-specific modification mapping. Most commonly used enzyme in LC-MS-based poly(A) tail analyses
RNase A	3'-end of pyrimidines (C, U) [40]	Oligonucleotides	Modification site determination in pyrimidine-rich regions
RNase U2	3'-end of purines (A, G) [41]	Oligonucleotides	Identifying modifications in purine-rich sequences
Cusatavin	cleave Cp[U/A/G] dinucleotide combinations along with UpA and [A/U]pU [42]	Oligonucleotides	Selective cleavage for cytidine modification analysis
MC1	cleave the [A/U/C]pU dinucleotide bond [42]	Oligonucleotides	Modification detection in uridine-rich sequences
RNase 4	Cleaves between UG and UA [84]	Oligonucleotides	Identification of various uridine nucleobase modifications
Colicin E5	Physiological target sequence is UQU (Q: queuosine) of tRNAs [33,103,104]	Oligonucleotides	Specifically cleaves QUN anticodons of the tRNAs for Tyr, His, Asn and Asp.
Nuclease P1	3'-end of single-stranded RNA [105]	generate intact 5' cap structures and NMPs or combined with additional enzymatic steps to generate the corresponding nucleosides	Identifying modifications in single-stranded RNA
Phosphodiesterase I	3'-end of nucleotides in single-stranded RNA [106]	Mononucleotides	Complements nuclease P1 digestion for complete hydrolysis
Bacterial Alkaline Phosphatase	Removes phosphate groups [106]	Free nucleosides	Prepares nucleosides for LC-MS analysis by eliminating charged groups

pairing reversed phase-LC analysis, lower-mass oligonucleotide sequences elute earlier, while higher-mass sequences elute later, resulting in a sigmoidal curve when retention time is plotted against mass. The method enables bidirectional sequencing, with its success relying on the correlation between the mass of formic acid hydrolyzed oligonucleotides and the chromatographic retention time. However, this approach requires paired-end reads from both the 5' end and the 3' end of RNA sequences to resolve terminal nucleotides and cannot reconstruct a complete sequence from a single ladder. To address these limitations, the two-dimensional hydrophobic end-labeling strategy introduces a hydrophobic tag at the 5' and/or 3' ends, significantly delaying the retention time of labeled oligonucleotide sequences [45]. This modification enhances the separation of the two mass ladders, enabling complete RNA sequence determination from a single ladder without

paired-end reading. The LC-MS data are analyzed using an algorithm implemented in Python that generates paired end sequencing reads from a set of compounds defined by their mass, chromatographic RT, and abundance. This method has been applied to the sequencing analysis of short synthetic RNAs (~20 nt) [45], tRNAs (~76–90 nt) [46], and a mixture of RNA oligonucleotides with a different sequence and/or modified nucleotides [47].

2.1.4. Stable isotope-labeled profiling of RNA modifications

Although RNase and acid digestions combined with high resolution MS analysis allows unambiguous determination of base composition of oligonucleotides, this approach is less suitable for quantitative analysis. Stable isotope labeling is a well-established method to evaluate metabolic activity in biological systems [48,49]. The incorporation of stable isotopic labels allows for the analysis of the dynamics of RNA structure and modification profile [50]. For example, Gameiro et al. quantitatively assessed the modifications in non-ribosomal RNAs in 786O cells using [¹³C-methyl]-methionine labeling. [¹³C-methyl]-methionine can be converted to [¹³C-methyl]-S-adenosylmethionine which is the direct substrate of RNA methylation reactions. This method can quantify the turnover of base modifications in newly transcribed RNA. The purified RNA was digested into ribonucleosides using nuclease P1 and analyzed by targeted LC-MS [51]. Taoka group reported a stable isotope-labeled approach for the comprehensive identification and quantification of RNA modifications. They incorporated an in vitro-transcribed, heavy isotope-labeled reference RNA into a sample RNA solution, digested the mixture with RNases and detected the post transcriptionally modified oligonucleotides quantitatively based on shifts in retention time and the MS signal in LC-HRMS. They mapped post-transcriptional modifications of the ribosome RNA (rRNA) from *Schizosaccharomyces pombe* [52] and *Saccharomyces cerevisiae* [53]. Additionally, RNA modifications in the human 80S ribosome were quantitatively analyzed using this approach [54]. Further optimization of the method led to a platform for quantitatively characterizing mRNA therapeutics, in which the primary structures of 200 to 4300 nt mRNAs including chemical modifications were confirmed [55].

2.2. LC-MS platforms and tandem MS techniques

Analysis of RNA modifications requires distinct LC-MS methodological and instrumental approaches. LC is critical in the analysis of RNA modifications, which reduces sample complexity by separating molecules and removing salts from formulation buffers prior to MS detection. Reversed phase ion pair liquid chromatography (IP-RPLC) remains predominant for RNA analysis. It could separate RNAs with sizes up to 8000 nt [56]. Hydrophilic interaction liquid chromatography (HILIC) is increasingly being developed and adopted [34,57,58], which captures degraded oligonucleotide down to a single nucleotide [27]. Micro- and nano-flow chromatography systems operated under size exclusion mode have recently been developed as effective on-line desalting platforms for the MS analysis of RNA, which produce high-quality full scan spectra with minimal salt adduction [28,59].

Mass spectrometry has become a critical tool in the characterization of modified RNAs. Recent reviews have comprehensively summarized the progress of MS in the analysis of nucleic acids [34,60]. RNA sequencing often relies on high-resolution mass spectrometry, such as time-of-flight (TOF), Orbitrap, and Fourier-transform ion cyclotron resonance (FT-ICR) mass spectrometers. These instruments provide the mass resolving power and accuracy needed to determine the exact *m/z* values of modifications. Optimization of tandem mass spectrometry (MS/MS) parameters utilizing collision-, electron- or photo-dissociated methods have enabled effective bond cleavage throughout the phosphodiester backbone while limiting secondary fragmentation, allowing characterization of modifications in larger (~100 nt) RNAs. High-sequence coverage can be achieved by reconstructing RNA sequences through the complementary c-/y-type ions and

non-complementary d-/w-type ions (Fig. 1).

Collision-activated dissociation (CAD), available in almost all mass spectrometers, generating sequence-informative a-/w-, b-/x-, c-/y-, and d-/z-type ions (Fig. 2). However, its performance is highly dependent on charge state, yielding optimal sequencing data for precursors in lower charge states [60]. Radical-based activation approaches, including ExD techniques and ultraviolet photodissociation (UVPD), utilize the interaction of electrons and photons with RNA to generate informative product ions [61]. These techniques are particularly advantageous because the labile P–O bond in a phosphate group remains stable in electron-based methods, while it is more fragile in collision-based methods [61]. For example, electron detachment dissociation (EDD), available on systems of FT-ICR and branched magneto-RF ion trap installed in the ZenoTOF 7600, produces d- and w-type product ions [62,63]. Negative electron transfer dissociation (NETD) on a modified Orbitrap Fusion Lumos can retain modifications of RNA while generating sequence-informative d- and w-type ions [64]. Ultraviolet photodissociation (UVPD) leverages UV light to induce rich, sequence-informative fragmentation of RNA [65]. It produces lower abundances of base-loss products common to CAD methods, and offers high sequence coverage and compatibility with long RNAs. Consequently, the most comprehensive characterization of large RNAs can be achieved not by a single method but by integrating data from multiple activation techniques, leveraging the complementary strengths of each to overcome their individual shortcomings.

3. Data processing tools

Mass spectrometry involving CAD has been the standard approach for the detection of modified nucleosides. Upon CAD, RNA preferentially generates c- and y-type ions [66]. Tools that allow complex full scan and MS/MS spectra to be processed in a (semi)automatic fashion are of great importance. Commercial software such as ProMass (Novatia) provides automated platforms for LC-MS data deconvolution. Other proprietary software including BioPharma Finder (Thermo Fisher Scientific), OligoQuest (Bruker Daltonics) and PMI-Byos Oligo (Protein Metrics) enable oligonucleotide impurity profiling and comprehensive mRNA mapping. A variety of freely available software programs have been developed, which offer distinct advantages for RNA modification analysis as reviewed recently [34,60]. Unlike commercial platforms, which often prioritize standardized workflows, open-source platforms grant researchers flexibility in parameter customization. This is critical for characterizing unconventional or newly discovered RNA modifications that may lack predefined settings in commercial packages [67,68]. Furthermore, the open-source tools rapidly integrate cutting-edge computational innovations, such as customizable search algorithms [69], advanced spectral matching techniques [70], and community-vetted statistical validation strategies [71]. Crucially, tools like OpenMS enable integration into extensible computational frameworks, empowering users to construct tailored workflows [72] (Table 2). Therefore, open-source tools are invaluable for collaborative research requiring full transparency in the rapidly advancing field of RNA modification analysis, whereas commercial software excels in regulated applications demanding vendor-supported compliance.

The first-generation and classical open-source software algorithms primarily focus on matching experimentally obtained product ion or oligonucleotide sequence data with computationally predicted data [34, 60]. The second-generation algorithms incorporate isotope envelope analysis, comparing the isotope distribution observed in an experimental spectrum with theoretically expected patterns calculated from elemental compositions to enhance matching accuracy and reliability [71,73,74]. These open-source software can be further categorized based on specific application scenarios and corresponding LC-MS analytical methods, as outlined below.

3.1. Top-down sequencing using collision-based dissociation

Rozenski and McCloskey developed the Simple Oligonucleotide Sequencer (SOS) in 2002 [75]. It facilitates the *ab initio* determination of nucleotide sequences based on mass spectra derived from CAD. However, SOS can only analyze oligonucleotides shorter than 20-mer and process a limited number of modifications at a time [76], which restricts its use for relative large RNAs with complex modifications. Therefore, it is more widely used in the analysis of enzyme digested RNAs which generate oligonucleotides that are suited to be analyzed by SOS [77]. Similarly, RoboOligo is an interactive program for the *de novo* sequencing of short (≤ 20 -mer) RNA oligomers [78], and Aom²S has been developed and applied to the analysis of a short RNA molecule with an inosine modification (ACGGCUUIGG) [73]. Oligonucleotide mass assembler (OMA) and oligonucleotide peak analyzer (OPA) calculate theoretical data and match with the experimental data [79]. OMA calculates all the theoretical CAD product ions and the corresponding m/z values for a given oligonucleotide sequence [27]. OPA matches the product ion data of oligonucleotides and their adducts from the experiment to the predicted m/z values. A key limitation of these software is the inability to profile unknown oligonucleotides (untargeted analysis), and this constraint underscores the need for improved techniques to characterize RNA modifications.

OligoDistiller is a recent developed software for the LC-MS analysis of synthetic oligonucleotides and their impurities [74]. This software annotates known (targeted) and profiles unknown oligonucleotides (untargeted), which is important in understanding critical quality attributes of oligonucleotide therapeutics. The targeted mode detection algorithm can perform MS data annotation via customer database searching (e.g., expected impurities or transformations) and theoretical product ions mapping from CAD spectra. For the analysis of unknown oligonucleotide species, the *Pointless* algorithm was implemented and a template-free pattern-searching approach was applied to detect isotope envelopes that can be oligonucleotide-related. For each unknown feature detected by this software, multiple evaluation metrics were provided to account for mass deviation, peak missingness, and shape similarity. The software includes over 75 major impurities and degradation products in the default template which has been published as an online resource [24].

3.2. Top-down sequencing analysis using radical-based dissociation

Peters-Clarke et al. introduced a novel implementation of NETD that allowed for activation of RNA anions with fluoranthene cations on a quadrupole-Orbitrap-linear ion trap hybrid mass spectrometer [64]. They developed a C# script based on the framework of the C# Mass Spectrometry Library (CSMSL, <https://github.com/dbaileychess/CSMSL>) to analyze the NETD fragmentation of a 2'-O-methyl modified hexanucleotide [64]. Additionally, OligoSeq, a C# script-based platform was developed for the analysis of phosphorothioate modified RNAs by using the NETD fragmentations [80]. It was found that NETD produces fewer base-loss product ions than collision-based methods, and it is especially useful in diagnosing phosphorothioate incorporation, producing diagnostic a- and z-type ions at PS linkage sites, beyond the standard d- and w-type ions.

Unlike CAD/HCD/NETD which often favors certain cleavage sites, UVPD uniformly dissociates RNA, improving *de novo* sequencing accuracy. Additionally, it enables sequencing of longer RNA strands (> 50 nt vs ~ 20 nt of other fragmentation approaches) with high confidence. Lanzillotti and Brodbelt recently developed Shiny Application for Fragment Assignment by Relative Isotopes (Nucleo-SAFARI), a R Shiny application used for the top-down UVPD MS/MS analysis of nucleic acids [81]. The tool generates precursor and product ion formulas, monoisotopic masses, and isotopic distributions (IsoSpecR), then matches theoretical distributions to MS centroided data within a user-defined ppm tolerance (default: 10 ppm). A product ion is validated

Table 2
The open-source tools for the MS data analysis of RNA modification.

Type	Data analysis tools	Main Function	Mass spectrometer used for software development	Fragmentation	Released Year	Software platform	Accessibility	Application
Top-down analysis	Simple Oligonucleotide Sequencer (SOS) OMA and OPA	ab initio sequencing	Micromass Quattro II and Micromass Q-TOF, direct infusion	a, a-B, c, w, y	2002	Delphi	Open source on request	DNA and RNA ($\leq$ 20-mer)
		Matching with the experimental MS/MS data	Thermo LTQ Orbitrap XL mass spectrometer		2013	Java	http://www.schuerch.dcb.unibe.ch/omaopa	DNA and RNA
	RoboOligo	Automated and manual de novo sequencing	Thermo LTQ-XL linear ion trap, CAD	a-B, c, w, y	2015	Visual Studio and Visual Basic	http://bearcatms.uc.edu/new/limbachgroup_publication	RNA ($\leq$ 20-mer)
	Aom ² S	Automated fragment assignment	Thermo LTQ Orbitrap Elite FT-MS instrument, NCE	a, b, c, d, w, x, y, z	2020	JavaScript	https://mstools.epfl.ch/am2s	an inosine-RNA (ACGGCUUGIGG)
	OligoDistiller	Analysis of complex oligonucleotides and impurities	Bruker TimsTOF Pro; CAD data were acquired with Thermo Fusion Lumos, HCD	a, b, c, d, w, x, y, z	2025	R	https://github.com/daniellyz/OligoDistiller	impurity analysis of chemically modified RNAs (19 nt)
	A C# script based on the design framework of the C# Mass Spectrometry Library OligoSeq	interpretation of MS/MS spectra for ion statistics and sequencing	Orbitrap Fusion Lumos, NETD	predominantly d- and w-type product ions	2021	C# script	https://github.com/dbaileychess/CSMSL	a 6-mer chemically modified- and a 21-mer oligonucleotides
	Nucleo-SAFARI	interpretation of MS/MS spectra for ion statistics and sequencing	Orbitrap Fusion Lumos, NETD/ CAD/HCD	a-, d-, w-, and z-type ions in NETD; c-, y-, b-, x-type ions in CAD and HCD	2024	C# script	https://github.com/coongroup/Transcriptomics	phosphorothioate modified RNAs (20-mer)
	FAST MS	Candidate sequence search	Orbitrap Lumos, UVPD/CAD	a, b, c, d, w, x, y, z	2024	R	https://github.com/mblanzillotti/Nucleo-SAFARI	DNA and RNA (10–80 nt)
Bottom-up analysis	Mongo Oligo Mass Calculator	Automated fragment assignment	apex-ultra hybrid Qq-FTMS (CAD, RTD, ECD, or EDD)	c and y ions in CAD	2024	Python	https://github.com/michael-palasser/FAST-MS/releases	15–47 nt RNAs
	Ariadne	Database search	N/A	N/A	1999	JavaScript	http://rna.rega.kuleuven.be/masspec/mongo.htm	
		Database search	Thermo LTQ-Orbitrap hybrid mass spectrometer and Waters Q-ToF Ultima mass spectrometer, CAD	a, c, w, y; up to 15 nt	2009	Web application	https://ariadne.riken.jp	RNA
	RNAmodMapper (RAMM)	Database search	Thermo LTQ-XL mass spectrometer and Waters Synapt G2-S HDMS mass spectrometer, CAD	a-B, c, w, y; up to 20 nt	2017	Java	http://bearcatms.uc.edu	RNA
	Nucleic Acid Search Engine (NASE)	Database search	Thermo Q Exactive HF mass spectrometer, CAD	a, b, c, d, w, x, y, z	2020	Implemented in C++ within the OpenMS framework	https://www.openms.de/nase	lncRNA, tRNA, rRNA
	Pytheas	Database search	Agilent Q-TOF 6520 ESI instrument, Waters Synapt G2-S system and Thermo Orbitrap Fusion Lumos instrument, CAD	a, b, c, d, w, x, y, z	2022	Python	https://github.com/ldascenzo/pytheas	tRNA, rRNA
	Nucleo's ID	Analyzing the completely digested nucleosides	Bruker MaXis 4 G mass spectrometer, CAD	ribose lost product ions of nucleoside	2023	Python	https://github.com/MSARN/NucleosID	RNA
	NuMo Finder	Analyzing the completely digested nucleosides (SWATH data)	Sciex ZenoTOF 7600 system, zenoSWATH	ribose/deoxyribose lost product ions of nucleoside	2023	Python	https://github.com/ChenfengZhao/NuMoFinder	RNA and DNA
Others	FeatureHunter	Analyzing the adduct-type modifications on RNA	Orbitrap Fusion Lumos		2024	C and C#	Provide the full pseudocode	DNA and RNA
	POINTLESS4DNA	Predict the isotope distribution	Synapt G2 HDMS QTOF mass spectrometer, full scan	N/A	2021	R	https://valkenborg-lab.shinyapps.io/pointless4dna/	DNA and RNA
	MIND4OLIGOS	Predict the monoisotopic mass	Bruker timsTOF Pro mass spectrometer, full scan	N/A	2024	R	https://valkenborg-lab.shinyapps.io/mind4oligos/	

if $\geq 70\%$ of its predicted isotopic probability matches observed peaks. In addition, hydrogen shifts may occur during bond cleavage induced by UVPD, influencing the isotopic distribution of fragment ions. Nucleo-SAFARI can also employ a non-negative least squares model to predict the relative amounts of the overlaid species generated by hydrogen shifts. This approach was applied to the characterization of tRNA-Phe methylations and a guanine to adenine mutation [28].

Free Analysis Software for Top-Down Mass Spectrometry (FAST MS) was developed to analyze linear polymers and their modifications, including positional isomers, while enabling relative quantification of site-specific changes [69]. The software is adaptable to different dissociation techniques (e.g., CAD, ECD/ETD, or UVPD). By leveraging the intact RNA sequence, FAST MS identifies, assigns, and relatively quantifies isotopic distributions in peak lists obtained from MS or MSⁿ spectra. Users can define monomer units, modifications, product ion types, and elemental isotope abundances, offering high flexibility for targeted top-down mass spectrometry analysis.

3.3. Data processing tools for the bottom-up analysis of RNA modifications

The analysis of digested RNA fragmentation data via MS/MS is a critical step in the bottom-up RNA sequencing, enabling the characterization of large RNA sequences and their modifications. Considerable efforts toward automating data analysis have been made in recent years, and classical data analysis tools take the given sequence of an oligonucleotide as input and calculate the theoretical m/z values of oligonucleotides as reviewed recently [34,60].

Due to the limited variability of RNA constituents (4 nucleotides versus 20 amino acids in proteins), RNase digestion is less likely to generate unique oligonucleotide sequences that can be unambiguously assigned to a single RNA species. To address this challenge, the following computational approach has been employed: (1) mapping all experimentally observed oligonucleotide sequences against database entries of known RNAs, (2) evaluating the matching of experimental sequences to reference RNAs using a probability-based scoring metric, and (3) identifying the RNA species with the highest consensus score as the most probable match. Four prominent tools have been developed for the bottom-up analysis of RNA modifications, including Ariadne [55, 82], RNAModMapper (RAMM) [83], NucleicAcidSearchEngine (NASE) [72], and Pytheas [71]. All four tools share a common workflow: (1) in silico digestion of user-submitted RNA sequences to generate theoretical oligonucleotide fragments, (2) matching experimental MS/MS spectra with these theoretical fragments, and (3) scoring and annotating matches to identify RNA sequences and modifications. They support customizable nuclease digestion and can handle a variety of RNA modifications, making them versatile for different experimental setups. However, their approaches and throughput differ significantly, leading to distinct applications and performance characteristics.

Ariadne is the first computational platform focusing on matching experimental spectra to theoretical ribonuclease-digested oligonucleotide sequences at scale. While Ariadne's two-step scoring algorithm (nucleotide fragment identification and nucleotide mapping) enhances specificity, the lack of large, established RNA databases and deficiencies in terms of statistical robustness limit its utility [82]. RAMM builds local databases from user-submitted sequences, accommodating over 120 post-transcriptional modifications and allowing synthetic modifications. It applies a dual scoring system, i.e., binomial P-score and dot product, to improve accuracy in product ion assignments. NASE and Pytheas, on the other hand, employ a target-decoy approach for false discovery rate (FDR) estimation, enhancing the confidence in modification identification.

Throughput and processing speed are critical for large-scale studies. NASE stands out in this regard, which can efficiently handle the analysis of RNA oligonucleotide data, especially of more complex samples or involving many different modifications in batch-compatible fashion. For

example, it processes a tRNA dataset with 23 modifications in <30 min, compared to an estimated month for RAMM and no results obtained in an equivalent search with Ariadne [72]. Pytheas, while offering robust statistical validation, lags in speed due to its comprehensive target-decoy approach, and one major limitation of Pytheas is the moderate speed of the search [71].

The choice of software for MS/MS data analysis of digested RNA fragmentation depends on the specific requirements of the study. For complex samples, NASE provides precursor ion mass correction and salt adduct analysis, enhancing its ability to identify longer oligonucleotides and complex modifications with unmatched speed and scalability [67, 68,84,85]. Pytheas supports the analysis of novel modifications, and it excels detailed modification mapping and statistical validation. Additionally, the flexible handling of isotope labeling is a key feature of Pytheas, which allows identification of pseudouridine in yeast 18S rRNA using metabolic ²H-labeling [71]. RAMM strikes a balance between flexibility and specificity, making it suitable for both predefined (~120) and user-defined RNA modification analyses [33,86,87]. Ariadne remains a reliable option for large-scale modification mapping despite its database limitations [88–90]. Taken together, these tools represent significant advancements in RNA modification analysis, each contributing unique strengths to the field.

3.4. LC-MS data analysis tools for the individual nucleotide units

Nucleos'ID is a user-friendly search engine designed for CE-MS/MS and LC-MS/MS data processing [37]. It matches the MS/MS data with a database containing m/z values of the precursor and product ions to identify modifications. Additionally, it enables untargeted searches, identifying both known and unexpected RNA modifications without prior assumptions. This is particularly valuable for discovering novel or unanticipated modifications in complex samples. While Nucleos'ID is a powerful tool for untargeted RNA modification identification, it can not discriminate isomers. NuMo Finder was developed to accommodate SWATH (sequential window acquisition of all theoretical fragment ion spectra) data in the analysis of individual nucleotide units [91]. Importantly, spectra derived from nucleobase fragmentation provide important information to unequivocally identify the positional isomers, and reference spectral libraries have been generated by employing homologous nucleobases to aid in identifying modified ribonucleosides and distinguishing them from their positional isomers when the standards are unavailable [92].

3.5. Other MS data processing tools for RNA modification analysis

3.5.1. Nucleic acid adductomic analysis

Exposure to physicochemical reagents may result in adduct-type modifications on RNA, which contains adducts on nucleosides and nucleobases, cross-links between nucleobases, RNA-RNA cross-links (RRCL) [93], and RNA-protein cross-links (RPCL) [94]. Adductomics is a rapidly evolving field with great potential to provide valuable insights into the mechanisms of chemical toxicity and disease. While LC-MS is the gold standard for detecting nucleic acid adducts, traditional workflows often require manual MS spectra inspection, limiting scalability. Wen et al. developed the FeatureHunter software to provide a comprehensive LC-HRMS data analysis workflow for nucleic acid adductomics [22]. This tool automates the extraction, annotation, and classification of various adducts, enabling more efficient nucleic acid adductomic analysis.

3.5.2. Predict the isotope distribution and monoisotopic mass

Identifying an oligonucleotide in a mass spectrum by comparing its observed isotope pattern with the theoretically expected pattern can be highly informative. However, this approach is limited when the molecular identity is unknown. To address this challenge, the POIN-TLESS4DNA tool was developed to detect and deconvolute nucleic acid

isotope patterns in mass spectra without requiring prior knowledge of elemental composition [95]. However, a critical drawback remains: the tool depends on the monoisotopic mass as input, which is often undetectable in RNA spectra due to its low abundance. To overcome this issue, MIND4OLIGOS was developed based on the MIND algorithm [96]. Instead of relying on the monoisotopic peak, it predicts the monoisotopic mass using the most abundant peak in the experimental spectrum. The predictions show strong agreement with actual oligonucleotide masses and probabilities, confirming the method's accuracy and reliability.

4. Future perspectives and conclusion

The rapid technological advancements in mass spectrometry (MS), coupled with growing interest from the pharmaceutical sector, have spurred new applications in nucleic acid research. However, progress in MS-based RNA analysis has lagged behind the explosive developments seen in proteomics. A unique challenge in RNA analysis is the limited diversity of nucleobases (A, C, G, U), which increases the prevalence of isomeric and repetitive sequences. Recent breakthroughs, such as cyclic ion mobility mass spectrometry, have enabled the distinction of oligonucleotide isomers [90,97], and improved product ion identification in top-down MS, as demonstrated by the analysis of a heavily modified 100-mer guide RNA [98]. Custom computational workflows (e.g., Python-based pipelines using pyOpenMS, SciPy, and NumPy) have been designed for analyzing the datasets.

The integration of artificial intelligence (AI) and machine learning (ML) is poised to revolutionize RNA modification research. RNA language pre-trained models leverage deep learning (e.g., Transformers) to capture intricate patterns in RNA biology [99]. AI/ML tools, such as those reviewed by Acera Mateos et al. [100], are already advancing RNA modification prediction, while innovations tools like LinearDesign optimize mRNA stability for vaccine development [101]. Moreover, modern ML methods and architectures have been applied in MS imaging and proteomic analyses [102]. Therefore, future efforts in RNA modification analysis should focus on merging AI/ML with MS-based workflows to enable intelligent spectral interpretation, automating the identification of modified residues and isomeric sequences. Additionally, AI/ML based prediction of RNase digestion patterns and LC retention times for oligonucleotide fragments will streamline the LC-MS method development. Furthermore, cloud-based and shared LC-MS database will promote the discovery and understanding of RNA modifications. Taken together, the convergence of new MS detection techniques, AI/ML, and cloud computing will be pivotal in unlocking the full potential of RNA modification studies, transforming LC-MS into a more robust, high-throughput, and accessible tool for nucleic acid research.

CRediT authorship contribution statement

Jiawei Wang: Writing – review & editing, Writing – original draft, Investigation. **Zhenwei Jiang:** Writing – review & editing, Writing – original draft, Investigation. **Zihui Jin:** Writing – original draft, Investigation. **Ziqiang Cao:** Writing – review & editing, Writing – original draft, Investigation. **Pan Deng:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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